

Assignment 3 : Machine Learning

Due Date: 27th April 2026

<https://www.kaggle.com/competitions/playground-series-s6e4/overview>

The link above is to an ongoing kaggle competition. The competition is part of the playground series especially designed to learn and test ML techniques. (Hint: It is a multiclass classification task.)

Here is what you need to do:

1. Signup to Kaggle, and join the competition above.
2. Train model including but not limited to Decision Trees, K-Means, Linear Regression as classifiers, Naive Bayes, etc.
3. Make your submissions in the competition.
4. Remember there are limited submissions per day allowed so make use of LOOCV and K-Fold

What to Submit:

1. submit your python file you used to train and test then finally train again on the whole train.csv with the best parameters and model selection before generating predictions for the test.csv file shared in the competitions.
2. Submit your final submitted csv file.
3. Submit a screenshot of the leaderboard the kaggle competition showing your name and score.
4. Submit a report mentioning which methods you used and accuracy you got with them. Along with confusion matrices of the failed attempts and what insight they gave you. A template is attached for reference you may make changes or write your own but make sure it included the content mentioned.

Judgement criteria

1. Your score in the leaderboard
2. Your approach
3. Your quality of code
4. Your reporting of failed attempts and final model Displaying Ass3.md.